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Rec A disruption unveils cross talk between DNA repair and membrane damage, efflux pump activity, biofilm formation in *Mycobacterium smegmatis*

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Running Title: RecA affects mycobacterial drug resistance mechanisms

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Abstract

Tuberculosis (TB) caused by *Mycobacterium tuberculosis* (MTB) has emerged in recent decades as one of the leading causes of mortality worldwide. The burden of TB is alarmingly high, with one third affected global population as reported by WHO. Short-course treatment with an antibiotic is a powerful weapon to treat infection of susceptible MTB strain, however; MTB has developed resistance to anti-TB drugs, which is an escalating global health crisis. Thus there is urgent need to identify new drug targets. RecA is a 38 kilodalton protein required for the repair and maintenance of DNA and regulation of the SOS response. The objective of this study is to understand the effect of disruption of RecA gene (deletion mutant Δ disA from previous study) in a surrogate model for MTB, *Mycobacterium smegmatis*. This study demonstrated that disruption of RecA causes enhanced susceptibility towards rifampicin and generation of ROS leading to lipid peroxidation and impaired membrane homeostasis as depicted by altered cell membrane permeability and efflux pump activity. Mass spectrometry based lipidomic analysis revealed decreased mycolic acid moieties, phosphatidylinositol mannosides (PIM), Phthiocerol dimycocerosate (DIM). Furthermore, biofilm formation was considerably reduced. Additionally, we have validated all the disrupted phenotypes by RT-PCR which showed a good correlation with the biochemical assays. Lastly, RecA mutant displayed reduced infectivity in *Caenorhabditis elegans* illustrating its vulnerability as antimycobacterial target. Together, present study establishes a link between DNA repair, drug efflux and biofilm formation and validates RecA as an effective drug target. Intricate studies are needed to further understand and exploit this therapeutic opportunity.

Keywords- *Mycobacterium*; Rec A; cell membrane; lipidomics; Biofilm; *C. elegans*.

1. Introduction

Despite the improvement of tuberculosis (TB) regimen, TB remains the second public severe health problem and a leading cause of death due to an infectious disease after HIV infection (Well et. al. 2007). The treatment of TB caused by *Mycobacterium tuberculosis* (MTB) requires at least six months duration, but multidrug-resistant MTB entails second-line drugs that are generally toxic for a longer period usage (Riekstina et. al 2007). Currently, the main obstacle in TB research is limited understanding of the mechanisms by which MTB evades both the host immune response and the emergence of multidrug resistance (MDR). Hence we need to identify novel drug targets to be employed in therapeutic strategies.

The RecA proteins are structurally conserved among eubacteria and archaea (Rocca et. al. 1997). However, the functional connection among RecA proteins is poorly implicit. In comparison to the most-studied *Escherichia coli* RecA, which is essential for several processes related to DNA metabolism, MTB RecA displays many unique, distinctive features (Ganesh et.al 2003; Movahedzadeh et. al. 1997). In vitro studies have revealed that the MTB RecA protein exhibits numerous differences from other bacterial RecA homologs. Apart from *E. coli* RecA (EcRecA), homologous pairing and strand exchange promoted by MTB RecA (MtRecA) is greatly dependent on the pH of the medium (Vaze & Muniyappa, 1999). The MtRecA protein also displays a reduced affinity for ATP and reduced efficiency of ATP hydrolysis (Kumar et al., 1996). First time (Frischkorn et al., 1998) function of the MTB RecA protein in DNA repair and integration of exogenous nucleic acids was demonstrated by its ability to fully restore the phenotype of *M. smegmatis* RecA mutants. RecA also has a regulatory function in response to DNA damage, due to the ability of nucleoprotein filament created on regions of single-stranded DNA to stimulate the cleavage of repressor protein LexA (Friedberg, et. al 1995; Little et. al

1992). The RecA gene of MTB is remarkable, considering it is expressed by two promoters that expresses DNA damage response by unusual mechanisms (Davis et. al. 2002). RecA also regulates SOS response which is globally responsible for DNA damage (Cox et. al. 2007). RecA has a role in recombinational DNA repair which is mainly essential for replication re-open of a replication fork after subsequent collapse. In MTB transcriptional fusion assays specify that the regulation of *ruvC* was independent of RecA, while its expression may be modulated by LexA binding to the SOS box that flaps the transcription start site (Lisa F et. al. 2009). In addition, *lacZ* transcriptional fusions to break down the DNA upstream of the MTB *recA* gene in order to set up regions of *recA* expression and to recognize the promoter elements (Gopaul et. al. 2003). A study in mycobacteria suggests that the Clp-gene regulator ClgR is involved in regulating the LexA/ RecA-independent pathway (Wang et al., 2011). MTB needs integral RecA to control ROS and RNI making by infected macrophage, and this ROS production is ERK1/2-dependent (Szulc-Kielbik et. al. 2015).

The previous study (Manikandan et. al. 2018) reported that both cyclic di-AMP and cyclic di-GMP bind particularly to the C-terminal motif of *Mycobacterium smegmatis* (MsRecA) and MTB RecA (MtRecA). Furthermore, cyclic di-AMP controls the new mode of negative regulation of RecA gene expression, DNA repair and homologous recombination in mycobacteria (Manikandan et. al. 2018). In this study, we aimed to add to the existing literature and understand the link between RecA gene disruption, cell membrane integrity, drug efflux and biofilm formation along with validation as a drug target in a surrogate model for MTB, *M. smegmatis*. Considering the fact that MTB pathogenesis is interplay of several unknown factors, this study establishes that functional DNA repair pathway mediated by RecA is indispensable

when one considers dissecting drug resistance mechanisms in MTB and finding novel drug targets.

2. Material and methods

All Media chemicals Middlebrook 7H9 broth, Middlebrook 7H10 agar, albumin/dextrose/catalase (ADC), oleic acid/albumin/ dextrose/catalase (OADC) supplements were purchased from BD Biosciences (USA). Tween-80, INT (iodonitrotetra-zolium), Isoniazid (INH) and DCFDA (dichlorodihydro-fluorescein diacetate) were obtained from Sigma Chemical Co. (St.Louis, MO, USA). Ethidium Bromide (EtBr), Dinitrophenol (2,4,DNP), Propidium iodide (PI), Methanol (CH₃OH), Triton X, crystal violet (CV), peptone, ferrozine, calcofluor white, p-dimethyl aminobenzaldehyde (2-DOG), Brain heart infusion (BHI) were purchased from Himedia (Mumbai, India). Dimethyl sulfoxide (DMSO), Potassium chloride (KCl), Sodium chloride (NaCl), di-Sodium Hydrogen Orthophosphate (Na₂HPO₄), Potassium Di-hydrogen Orthophosphate (KH₂PO₄), Glycerol, magnesium chloride (MgCl₂), calcium chloride (CaCl₂), Sodium Dodecyl Sulphate (SDS) and ascorbic acid (AA) were obtained from Fischer Scientific. Agarose was purchased from CDH India. Sodium acetate was purchased from Qualigens Fine Chemicals (Mumbai, India).

2.1 Strains and growth culture

The *M. smegmatis* wild type (WT) and Δ disA (RecA mutant) strains used in this study were kindly provided by Manikandan et. al. (2018) who generated the mutant strain. This mutant was generated from *M. smegmatis* cells with the plasmid pYS1, a mycobacterial shuttle vector with a temperature-sensitive mycobacterial origin of replication and counter-selectable marker sacB. The 5' flanking sequence amplified with primers C89 FDisA and C90 RDisA is 713 bp long which includes 139 bp of the open reading frame (ORF) and the 3' homologous sequence

amplified with primers C91 FDisA and C92 RDisA is 818 bp long that contains 411 bp of the ORF. The construct deleted 570 bp from the ORF of *disA*. To create unmarked deletion of the gene, the *disA* marked deletion mutant cells were transformed with pML2714. pML2714 contains P1 Cre recombinase which recombines the loxP sites present on either side of gfp-hyg cassette resulting in loss of the cassette to create unmarked deletion mutant. *M. smegmatis* was grown in Middlebrook 7H9 (BD Biosciences) broth supplemented with 0.05% Tween-80 (Sigma–Aldrich), 10% ADC (BD Biosciences) and 0.2% glycerol (ThermoFischer Scientific) in 100-mL flasks and incubated at 37 °C. Cultures were subsequently grown on Middlebrook 7H10 (BD Biosciences) agar media supplemented with 10% (v/v) OADC (BD Biosciences) for solid agar allowing growth for 48 h at 37 °C. Stock cultures of log-phase cells were maintained in 30% glycerol and stored at -80 °C.

2.2 Minimum inhibitory concentration

Minimum inhibitory concentration (MIC) was determined using the broth-dilution method described previously (Hans et. al, 2017) according to Clinical and Laboratory Standards Institute guidelines. Briefly, 100 µL of Middlebrook 7H9 broth was placed in each well of a 96-well plate, followed by the addition of the drug with the remaining media and subsequent serially diluted 1:2. Cell suspension of WT and Δ *disA* (100 µL in normal saline at an optical density (O.D₆₀₀ nm) of 0.1) were added to each well, followed by incubation at 37 °C for 48 h. The MIC (Genetix, Biotech Asia Pvt. Ltd.) values were evaluated by observing the O.D₆₀₀ nm in a micro plate reader. The MIC was considered as the value which showed at least 90% inhibition in O.D₆₀₀ nm compared to positive control without any drug.

2.3 ROS estimation

WT and Δ disA cells were grown in middlebrook 7H9 broth until it reaches 0.8–0.9 O.D₆₀₀ nm. The cells were harvested at 13,000 rpm for 15 min and resuspended in 1 mL of media and grown for 1 h at 37 °C (Starvation Stress) with shaking. Cells were harvested, washed with PBS pH 7.4 and resuspended in same buffer in such a way that each 1 mL of PBS pH 8.5 buffers contains 3×10^7 cells. DCFDA was added in cell suspension with the final concentration of 10 mM and incubated at 37 °C for 30 min with continuous shaking. After washing with PBS buffer pH of 7.4 and resuspension in the same buffer the cells were observed in fluorescence microscope with the excitation and emission wavelength at O.D₄₈₈ nm, slit 5 nm and O.D₅₄₀ nm, slit 10 nm respectively at 40x magnification. 3 mM H₂O₂ was used as positive control (Miller et. al. 2010; Hans et. al. 2017). To quantify for the ROS generation, iodonitrotetrazolium chloride (INT) assay was performed as described previously (Pandey et. al. 2015; Hans et. al. 2017). 0.5 mL of 0.2% INT dye was added in each sample. All the tubes were kept at ambient temperature for couple of hours (i.e., 2–3 h) to check the color change in the tubes. INT dye is pale yellow in color, but after it reacts with ROS, the color is changed from pale yellow to dark pink color which was measured spectrophotometrically within 40 min in UV–vis Spectrophotometer at O.D₄₉₀ nm.

2.4 Estimation of MDA for lipid peroxidation

Estimation of MDA in the WT and Δ disA cells was done by thiobarbituric acid method with slight modification (Kulkarni et. al 2013). Thiobarbituric acid reacts with malondialdehyde (MDA) which is produced by hydrolysis of lipid hydroperoxides to form a pink red color complex that can be measured spectrophotometrically at O.D₅₃₂ nm. The results were expressed as nmoles/mL. This complex is usually quantified against MDA standards which are generated from 1, 1, 3, 3 tetraethoxypropane under the same reaction conditions.

2.5 Spot assay

Spot assays for the WT and Δ disA strains were determined using a method described previously (Ansari et. al. 2016; Hans et. al. 2019). Briefly, for the spot assay 5 μ L of fivefold serial dilutions of each WT and Δ disA culture (each with cells suspended in normal saline to an O.D₆₀₀ nm of 0.1) was spotted onto Middlebrook 7H10 agar plates. Growth difference was measured after incubation at 37 °C for 48 h.

2.6 Colony morphology and sliding motility

Colony morphology and sliding motility were determined as described previously (Pal et. al. 2016). Briefly, WT and Δ disA cells were plated on MB7H10 agar plates supplemented with 10% OADC (BD Difco) and incubated at 37 °C for 2 to 4 days. Post incubation, images of the individual colonies were taken at 10x magnification. To score for sliding motility, WT and Δ disA cells were grown until stationary phase in MB7H9 medium supplemented with 10% OADC. 3 μ L of the culture was spotted in the middle of MB7H10 plates solidified with 0.3% agarose without any carbon source. The plates were incubated at 37 °C for 4 days. Post incubation, images were taken at 10x magnification.

2.7 Cell sedimentation assay

Cell sedimentation assay was performed as described previously (Pal et. al. 2016; Hans et. al. 2017). Cultures of the WT and Δ disA at O.D₆₀₀ 1.0–1.4 in Middlebrook media supplemented with ADC were adjusted in triplicate to O.D₆₀₀ 1.0 and kept unshaken at 37 °C. At 3h and 22 h, the upper 1 mL was removed for O.D₆₀₀ nm measurements and graph was plotted as described in figure legends.

2.8 Propidium iodide uptake assay

Propidium iodide (PI) uptake assay was performed as described elsewhere (Shi et. al. 2007; Hans et. al. 2017). Briefly, WT and Δ disA cells were grown at 37 °C for 24 h in Middlebrook broth supplemented with 10% ADC. The cells were harvested by centrifugation and suspended in PBS to 0.5 OD₆₀₀ nm. 50 mg PI added to 1 mL bacterial suspension was placed in each tube and kept at room temperature for 15 min in the dark. 10 μ L bacterial suspension was transferred to a glass slide, covered with cover-slip and examined under fluorescence microscope at 40x magnifications.

2.9 EtBr efflux assay

The efflux of EtBr was determined by using protocol described previously (Sharma et. al. 2017). Briefly, approximately 1×10^6 WT and Δ disA cells were incubated until exponential phase. Cells were pelleted, washed twice with phosphate-buffered saline (PBS) (without glucose), and resuspended as a 2% cell suspension. The cells were then de-energized with an efflux pump inhibitor 2, 4 DNP (20 mg/mL) in PBS (without glucose). The de-energized cells were pelleted, washed, and then resuspended as a 2% cell suspension (w/v) in PBS without glucose, to which EtBr was added at a final concentration of 4 μ g/mL and incubated for 45 min at 30 °C. The equilibrated cells with EtBr were then washed and resuspended in such a way so as to form 2% cell suspension (w/v) in PBS (with glucose 0.4%) for EtBr efflux. Samples with a volume of 2 mL were withdrawn at different time interval and centrifuged at 10,000 rpm for 1 min. The supernatant was collected, and absorbance was measured at O.D₂₈₅ nm. Glucose-free controls were included in all the experiments.

2.10 Extraction of total lipid

Total lipids from WT and Δ disA were extracted by Folch Method with slight modifications (Pal et al., 2018; Sharma et. al. 2017). Cells grown upto exponential phase with 1.2 O.D₆₀₀ nm were used for lipid extraction. Cells were harvested at 10,000 rpm for 10 min. Cells were homogenized in aqueous solution for 3 min and suspended in CHCl₃ and CH₃OH in ratio of (1:2). Cells were then shaken well and centrifuged at 2000 rpm at 4 °C for 10–15 min. Supernatant was transferred to another glass vial and remaining CHCl₃ was added and filtered through Whatman No. 1 filter paper. The extract was then washed with 0.88% KCl to remove the non lipid contamination. Two layers were formed but the lower dense layer of CHCl₃ containing lipid was taken by glass Pasteur pipette in 5 mL glass vial having Teflon capping. The vials are stored at -20 °C until further analysis.

2.11 Extraction of cell wall lipid

The cell wall lipids from WT and Δ disA strains were isolated as described previously (Singh et al. 2012; Pal et al., 2018). WT and Δ disA cultures were grown in Middlebrook 7H9 supplement with ADC. The cultures were pelleted down and ~ 2 grams of mycobacterial cell pellet was extracted in 1mL hexane at room temperature overnight. The hexane layer was then transferred to 1:1 CHCl₃ / CH₃OH and was analyzed for the free mycolic acids.

2.12 Ultra performance liquid chromatography electrospray ionization and mass spectrometry (UPLC-ESI-MS)

The lipid analysis of WT and Δ disA lipid samples were performed on reverse-phase ultrahigh-pressure liquid chromatography (UHPLC, Exion LC Sciex, USA) using an Kinetex C18, 2.1 × 50 mm column (Phenomenex, USA) with a particle size of 1.7 μ m column oven kept at 60 °C,

coupled to a hybrid triple quadrupole/linear ion trap mass spectrometer (4500 Q-TRAP, SCIEX, USA) and elution was done for 30 min, using mobile phase A and B. Solvent A have 10% methanol and 90% buffer of ammonium acetate (5mM) dissolved in 95:5 water and acetonitrile and solvent B have 5% isopropanol, 10% methanol and 85% acetonitrile. A linear gradient elution was performed at the flow rate of 0.250 mL min⁻¹ as 30-50% B for 0-5 min; 50-70% B for 5-8 min; 70-95% B for 8-23 min; 95-30% B for 23-25 min; and 30% B up to 30 min. The ion Spray heater was maintained at 500 °C, with ion Spray needle voltage set to 5500 V, curtain gas was set to 30 psi, and GS1 & GS2 were set to 50 psi, declustering potential (DP): 200 V and entrance potential (EP) 10 V. The sample was introduced using an autosampler with 5 µL of sample injection volume. The data were recorded in the mass range, m/z 200–2000 Da in ESI± mode and processing was performed with the Analyst software, where each chromatogram was smoothened and the background was subtracted. The analysis of the acquired data was performed by mass spectrometry-based lipid (ome) Analyzer and molecular platform (MS-LAMP) software (Sabareesh and Singh 2013; Layre et al. 2014; Pal et al. 2018). This software is a graphical user interface (GUI) standalone programme built using Perl::Tk. It is a combination of MTB lipid database (www.mrl.colostate.edu) and lipid metabolites and pathways strategy consortium (LIPID MAPS; www.lipidmaps.org). The adduct used for analysis were [M + H]⁺, [M + NH₄]⁺, [M + Na]⁺, [M - H]⁻ and 1.0 window range was used for analysis. Further, it needs to be noted that the data herein are analyzed qualitatively only. All the experiments were performed in triplicates to ensure reproducibility and accuracy.

2.13 Biofilm formation

M. smegmatis biofilm forming potential was both qualitatively and quantitatively analyzed using the microtiter-plate method as described previously (Hans et. al 2017; Sharma et. al. 2016). WT

and Δ disA cultures were grown overnight at 37 °C in Middlebrook media, followed by transfer of 100 μ L media in each well of the 96-well plate. Cultures that had reached an OD₆₀₀ nm of 0.1 were diluted (1:100) using Middlebrook media, and 100 μ L of each diluted culture was pipetted into each well of 96-well flat-bottom microtiter plate and incubated at 37 °C for 48 h. The wells were rinsed with water, and 125 μ L of 0.1% solution of the crystal violet was added (Trivedi et.al. 2016). Plates were incubated for 10 min, followed by washing with distilled water three times and observed under a light microscope at 100 x magnifications. For quantitative assays of the biofilm, 200 μ L of 95% ethanol was added to each crystal violet stained well, and plates were incubated for 10 min at room temperature. Contents of each well were mixed by pipetting, followed by transfer of 125 μ L of the crystal violet/ethanol solution from each well to a separate well of an optically clear flat bottom 96-well plate. The O.D₆₀₀ nm was measured using a spectrophotometer. The inhibition of biofilm in Δ disA was expressed as percentage of inhibition/reduction. Biofilm biomass was measured as described previously (Hans et. al. 2017; Stiefe et. al. 2016). Pre-weighed sterile square coverslip was used. Exponentially grown cells Δ disA and WT were diluted to an O.D₆₀₀ nm of 0.1 with Middlebrook 7H9 broth and the suspension was added to a sterile 12-well plate with coverslip in each well. The inoculated plate was incubated at 37 °C for 90 min with gentle agitation (150 rpm) until adhesion occurred. The coverslip were washed with PBS and then moved to a new 12-well plate containing 2 mL of fresh Middlebrook 7H9 broth. The plate was incubated at 37 °C for an additional 72 h with agitation at 75 rpm to allow biofilm formation. For dry mass measurements, the total biomass of each biofilm was calculated by subtracting the pre-determined weight of the silicon squares.

2.14 RT-PCR

RNA was extracted from WT and Δ disA cells by standard TRIzol protocol as described previously (Pal et al. 2019). Briefly, 500 μ L of TRI reagent (trizol) was added to WT and Δ disA cells and sonicated. Reverse transcriptase (RT) PCR was done as described in the RevertAid H Minus kit (Invitrogen). 5 μ g isolated RNA was DNase treated at 37 °C for 30 min and reaction was terminated by adding 1 μ L of 25 mM EDTA and incubated at 65 °C for 60 min. RNA was subsequently primed with oligo (dT) 18 for cDNA synthesis at 42 °C for 60 min. Reverse transcription reaction was terminated by heating at 70 °C for 5 min. The synthesized cDNA product (2 μ L) was directly used for PCR amplification (50 μ L) using gene-specific forward and reverse primers (Table S1). The amplified products were gel electrophoresed and the densities of bands (for genes of interest) were measured using Image J software.

2.15 Maintenance and survival of *C. elegans*

Worms were grown on NGM plate at 20 °C, unless otherwise indicated, and routinely maintained on *E. coli* OP50. *C. elegans* were routinely purified by the alkaline hypochlorite method. *C. elegans* N2 was maintained and propagated on *E.coli* strain OP50 with standard technique. Approximately 2 to 4 *C. elegans* of L4 stage and adult hermaphrodites were transferred from a lawn of *E. coli* OP50 to another plate for maintenance. For the *C. elegans* co-infection liquid assay, a methodology that was described previously was used (Hae et. al. 2017; Pal et. al. 2019). Briefly, synchronized, young adult nematodes were sequentially preinfected with Δ disA and WT culture of *M. smegmatis* for 2 h on BHI agar medium containing ampicillin (45 mg/mL). The nematodes were then washed four times with 2mL of sterile M9 buffer. They were collected by centrifugation at 900rpm between each wash and before being pipette (30 worms per well) into one well of 12 well microtiter dish containing 2mL liquid medium. The infected *C. elegans* was directly inoculated into the liquid medium for feeding of 5 days plates and worms were

considered dead when they did not respond to the tapping on the plate and scored as live or dead on a daily basis on the 5th day of infection.

3. Results

3.1 Rifampicin susceptibility is enhanced in Δ disA

Drug susceptibility testing was performed by broth microdilution assay as described in methods. We observed that MIC for WT towards rifampicin (RIF) was 0.25 μ g/mL (Fig. 1). Interestingly, RIF susceptibility was enhanced to 62.5ng/mL in Δ disA (Fig. 1). Of note there was no growth defect in Δ disA as compared to WT (Fig. S1).

3.2 Endogenous ROS levels are enhanced in Δ disA

To confirm the effect of RecA disruption on the redox status of mycobacterial cell, we measured the production of ROS in Δ disA using a fluorescent reporter (DCFDA). We found increased fluorescence in Δ disA in contrast to WT (Fig. 2a). This observation was further validated by quantifying the ROS production using INT dye as described in methods. We observed increased pink color in Δ disA cells in contrast to WT confirming enhanced ROS generation comparable to H₂O₂ (positive control). Interestingly, the increased ROS production could be reverted by addition of AA (ascorbic acid), a known antioxidant (Fig. 2b). Thus we inferred that ROS is enhanced upon RecA disruption.

3.3 Lipid peroxidation is induced in Δ disA

Next, we studied lipid peroxidation which is a hallmark of enhanced ROS generation. We estimated lipid peroxidation, by determining the MDA levels in Δ disA and WT. MDA is used as

an oxidative biomarker for lipid peroxidation whose production is identified by measurement of reaction between MDA and TBA. Interestingly, we found enhanced MDA levels in Δ disA in comparison to WT (Fig. 2c) which confirmed the effect of RecA disruption on lipid peroxidation.

3.4 Membrane integrity is disrupted in Δ disA

Next, we sought to elucidate the repercussion of lipid peroxidation on membrane damage. Firstly, we performed a spot assay with known membrane perturbing agent SDS. We found that Δ disA was hypersensitive in the presence of SDS that indicated disrupted membrane homeostasis (Fig. 3a). Further, we examined the PI influx, where enhanced fluorescence intensity confirmed membrane perturbation in Δ disA, contrary to WT (Fig. 3b). Additionally, we studied membrane permeability of Δ disA by nitrocefin hydrolysis. We observed that membrane permeability was considerably enhanced in Δ disA as compared to WT (Fig. 3c). All these observations confirmed that membrane integrity was disrupted upon RecA deletion.

3.5 Cell surface phenotypes are altered in Δ disA

Taking clue from the above observations, we further studied the cell surface phenotypes more closely. Firstly, we observed that colony morphology was altered in Δ disA displaying a rough and dry bordered colony in comparison to WT, showing smooth and waxy borders (Fig. 3d). Then we studied sliding motility which was inhibited in Δ disA in contrast to WT (Fig. 3e). Furthermore, we estimated the cell sedimentation rate of Δ disA. We observed that the rate of sedimentation was considerably enhanced in Δ disA as compared to WT (Fig. 3f). Notably, the difference was only pronounced after 22hours. All these observations suggested that deletion of RecA impaired the cell surface phenotypes.

3.6 Efflux pump activity is impeded in Δ disA

Next, we examined the effect of RecA deletion on efflux pump activity by monitoring the EtBr efflux. We observed inhibited efflux pump activity in Δ disA in contrast to WT as estimated by the extracellular EtBr concentration of supernatant (Fig. 3g). This suggested that RecA deletion has affected the capability of mycobacterial cells to efflux EtBr mediated by efflux pumps on the mycobacterial membrane.

3.7 Global lipidome of total lipids in WT and Δ disA

Comparative lipidome profile of WT and Δ disA was performed by mass spectrometry based lipidomics technology as described in methods. Fig. S2-S3 depicted representative chromatogram and mass spectra of WT and Δ disA. We analyzed our LC-MS data with window range (WR) of 1.0 and the data of total lipid revealed that 45 m/z of WT codes for 72 lipids and 40m/z encodes 68 lipids in case of Δ disA (Fig. 4a). The Venn diagram displayed the number of lipids that were specific and common between WT and Δ disA (Fig 4b). Out of 72 potential lipids interpreted from WT, 23 belonged to the Fatty acids (FA) category, 30 are Glycerolipids (GL), and 18 are Glycerophospholipids (GPL) (Fig.4c). On the other hand, the 68 prospective lipids found from the Δ disA strain comprised 26 FAs, 24 GLs, and 17 GPLs (Fig. 4c).

3.8 Alteration of lipid profile in Δ disA

We found decreased mycocerosic acid and mycocenoic acid in Δ disA. Besides these, we also found the absence of Alpha MA, Methoxy MA and DIMB in Δ disA (Fig. 4d). Further, we found decreased MG and TG in Δ disA as compared to WT (Fig. 4e). Similarly, while analyzing another important class of lipid, GPL, we observed the absence of PIM moieties in Δ disA viz.

PIM1, PIM2, lysoPIM3 and Ac1PIM1 (Fig. 4f). Thus, the output from MS-LAMP obtained upon querying LC-ESI-MS data acquired from both the WT and Δ disA suggested that the deletion of RecA affected all three major classes of lipids, although the effect was more pronounced for subclass FA.

3.9 Alteration of cell wall lipid profile in Δ disA

Fig. S4-S5 showed representative chromatogram and mass spectra of WT and Δ disA. We analyzed our LC-MS data with window range (WR) of 1.0 and the data of cell wall lipid revealed that 34m/z of WT codes for 53 lipids and 31m/z encodes 54 lipids in case of Δ disA (Fig. 5a). The Venn diagram displayed the number of lipids that were specific and common between WT and Δ disA (Fig. 5b). Out of 53 potential lipids interpreted from WT, 25 belonged to the FAs category, 20 are GLs, and 15 are GPLs (Fig. 5c). On the other hand, the 54 prospective lipids found from the Δ disA strain comprise 12 FAs, 22 GLs, and 16 GPLs (Fig. 5c). We observed a decrease in phthioceranic acid, hydroxy phthioceranic acid, mycocerosic acid and mycosanoic acid in Δ disA (Fig. 5da). TG was enhanced in Δ disA in comparison to WT (Fig. 5e). Similarly, we could observe higher PI, PE, and PG while analyzing GPL (Fig. 5f).

3.10 Analysis of LC-MS data using MycoMass database

We also analyzed the LC-MS data with another database, MycoMass reported by Layre et al. (2011), which contains 5399 molecules, nearly twice the number of lipids present in Mtb LipidDB (2518 lipids) reported by Sartain et al. (2011). As expected, the number of prospective lipids identified from MycoMass is greater than those resulted from Mtb LipidDB, simply because MycoMass contains nearly two-fold more lipids than Mtb LipidDB; however, of notable

importance we could notice that the trend of lipid class alterations in Δ disA was similar (Fig. S6).

3.11 Biofilm formation is inhibited in Δ disA

To study the effect of RecA deletion on biofilm formation, we performed qualitative assay where biofilms were visualized by CV staining. We observed that biofilm formation was considerably inhibited in the Δ disA in comparison to the WT (Fig. 6a). This result was further confirmed by quantitative methods by estimating the metabolic activity and biomass of biofilms. Both the assays established that metabolic activity and dry weight of biofilm biomass were considerably reduced by 70% (Fig. 6b) and 58% (Fig. 6c) respectively in Δ disA as compared to WT.

3.12 RT-PCR validates biochemical phenotypes in Δ disA

For confirmation of the above phenotypes, RT-PCR under similar growth conditions, as used for all biochemical assays, was performed. We selected about five genes from different phenotypic categories that were affected in Δ disA which included mycolic acid (*accd4*), cell membrane (*aftd*), ROS (*kat g*), DNA repair (*lig a*) and biofilm (*pe*). We found that RT-PCR showed a good correlation with abrogated phenotypes in Δ disA (Fig.7).

3.13 *C. elegans* infectivity is reduced with Δ disA

C. elegans, a widely used nematode model, was scrutinized to evaluate the survival in Δ disA infected nematode. We explored that the survival rate of infected worms were increased under infection with Δ disA as compared to WT establishing that RecA deletion led to a loss of infectivity (Fig. 8).

4. Discussion

MTB is a human pathogen that has staggering worldwide impact due to the emergence of MDR-TB against current therapeutics. A lot of the focus thus remains diverted towards the identification of novel drug targets. The protein RecA has a regulatory function for many processes associated to DNA damage and has been best characterized in *E. coli*. This study aimed to elucidate the effect of RecA deletion on MDR and virulence mechanisms of MTB in a surrogate model, *M. smegmatis*. Before conducting any experiment, we needed to rule out the concern of any growth defect associated with Δ disA. For this growth curve was performed and we observed that both Δ disA and WT grew at similar pace (Fig. S1).

RecA is a 38 kilodalton protein required for the repair and maintenance of DNA and regulation of the SOS response (Maslowska et al. 2019). The c-di-AMP-binding motif is conserved in mycobacterial RecA proteins. Mycobacterial RecA proteins port the c-di-AMP binding pocket and C-terminus of RecA is the section linked with regulation of its several functional activities (Manikandan et al. 2018). Moreover, RIF is known to target DNA-dependent RNA synthesis by inhibiting the β -subunit of DNA-dependent RNA polymerase encoded by *rpoB* gene and that over expression of *rpoB* gene makes MTB resistant to RIF (Telenti et al. 1993; Meier et al. 1996). Hence we were curious to observe the RIF susceptibility in Δ disA which was considerably enhanced as compared to WT (Fig. 1). RecA deletion mutant, which alters the capacity of MTB to repair DNA damage, and as consequence results in induction of different signaling response. Hence, the witness of intact RecA for MTB for control of nitric oxide and ROS production may be linked with the direct action of RecA in double strand break repair (Szulc-Kielbik et al. 2015). MTB utilizes a variety of pathways to protect against ROS by direct scavenging of the reactive species and induce DNA repair and protection of proteins (Nita et al.

2016). Thus we started our investigation by studying the redox status of Δ disA and found that the generation of ROS in Δ disA as compared to WT was significantly enhanced (Fig. 2a and b). ROS formation can lead to the development of chemical reactions causing damage to cellular components, including nucleic acids, proteins and lipids (Nita et. al. 2016). Lipids play a crucial role during the infection and they may be responsible for cell membrane damage which is caused by lipid peroxidation. Hence, we looked at one of the consequence of enhanced ROS for which we studied lipid peroxidation. Interestingly, we explored that lipid peroxidation was pronounced in Δ disA (Fig. 2c).

The significant phenotype that is observed in cells undergoing lipid peroxidation is membrane damage. Moreover, alteration in cell envelop properties and over expression of drug, efflux pumps activity are among the prominent mechanisms of MDR (Li et. al. 2009). Thus it was pertinent to study membrane structure and function more intricately in Δ disA. Several observations such as hypersensitivity in the presence of membrane perturbing agent (Fig. 3a), enhanced PI uptake (Fig. 3b) and nitrocefin hydrolysis (Fig. 3c) confirmed that cell membrane in Δ disA is more permeable and vulnerable. *Mycobacterium* contains unique cell wall architecture which makes it impermeable to anti-TB drugs (Marrakchi et. al. 2014). Furthermore, the cell surface phenotypes viz. colony morphology, sliding motility and cell sedimentation rate were altered in Δ disA reinforcing to hypothesize disruption in cell membrane (Fig. 3). Additionally, we observed that EtBr mediated efflux activity was also impaired in Δ disA (Fig. 3g). Active efflux of drugs governed by efflux pumps confers drug resistance developed by bacteria to counter the adverse effects of antibiotics (Li et. al. 2009). Hence the fact that membrane structure and function was disrupted in Δ disA observed in this study could be of clinical relevance.

Mycobacterium has distinctive cell-surface architectures comprised of peptidoglycan close to arabinogalactan, which is covalently linked to mycolic acid (MA). The MA is the most essential component of mycobacterial cell wall and striking target of numerous anti-TB drugs. The mycobacterium membrane is unique among Gram positive bacteria in synthesizing very-long chain fatty acids (FA) and MA, which functions as a permeability barrier. Hence, we performed mass spectrometry based lipidomics to comprehend the changes associated with the alterations in cell envelop of Δ disA. The mycobacterial cell wall is composed of complex lipids that are important for pathogenesis. The biosynthesis of FA plays a vital role in the formation of cell wall components, particularly MA which is required for the survival of MTB in the host (Kinsella et al. 2003). MAs are α - alkyl, β hydroxy long-chain FA found in abundance in the cell envelope of *Mycobacterium* which forms an efficient permeability barrier, modulates host innate immune responses and targets of several anti-TB drugs (Portevin et al. 2014). MTB contains three major classes of MAs: (1) alpha-, (2) methoxy- and (3) keto-mycolates (Watanabe et al. 2001). Mycolipanic acids are trimethyl branched FA present in the polyacrylate trehalose (PAT), and synthesized on the periplasmic side of the membrane and transported to the outer membrane of *Mycobacterium* that plays a structural role in the cell envelope as well as modulating host immune responses (Belardinelli et al. 2014). In our study, we observed decreased alpha, keto, methoxy MA in Δ disA hence confirming that the deletion of RecA leads to disrupt the synthesis of MA pathway of *M. smegmatis* which could be the causal reason for increased permeability of membrane (Fig. 4d). Increased mycolipanic acid and mycolipenic acid in Δ disA in case of total lipid profile also suggested that RecA deletion may lead to accumulation of mycolipanic acid inside the cell but doesn't get transferred outside the membrane of *Mycobacterium* which may be the reason that mycolipanic acid was not detected in the cell wall lipids.

GL is another class of lipids that contains three subclasses mono glycerols (MG), diacylglycerol (DG), and triacylglycerol (TG). TG is synthesized by three different enzymatic reactions: diacylglycerol: acyltransferase, phospholipid: diacylglycerol acyltransferase, and diacylglycerol: diacylglycerol acyltransferase and these three reactions play an important role in the acylation of DG, making this lipid a fundamental precursor of TG biosynthesis (Comba et al. 2013). TG is the primary form of storage lipid-based energy reserves in *Mycobacterium* which is catabolized by lipase enzymes in stressed condition and play a vital role in pathogenesis (Pal et al. 2019; Deb et al. 2006). Our data from total lipidome showed decreased TG in (Fig. 4e). Glycerophospholipids (GPL) is the significant component of cell membrane consisting of a molecule of glycerol which is attached with two FA, a phosphate, and usually one other small molecule (X). The name of GPL is derived from attached small molecules. Phosphatidylethanolamine (PE), phosphatidylinositol (PI), phosphatidylglycerol (PG), phosphatidylserine (PS) cardiolipin (CL), and mannosylated forms of PI known collectively as PIMs are the major structural lipid constituents of the plasma membrane of MTB (Jackson et al. 2014). Earlier studies have shown that *M. smegmatis* contain 37, 32 and 28% of CL, PE and PI/PIMs respectively out of the total phospholipid. Moreover, deletion of PI biosynthetic gene has been reported to be lethal in *M. smegmatis* suggesting that PI and glycosylated PI is very crucial for the survival for *Mycobacterium* (Jackson et al. 2014). PG involved in synthesis of CL, is one of the abundant phospholipids in *Mycobacteria* that form aggregates within membrane bilayer. PE is the structural component of plasma membrane whereas PI is the important class of phospholipids that are known to be further modified by extensive glycosylation (PIM, LM, LAM) (Sharma et al. 2017). PIM plays a crucial role in permeability of cell envelope, regulation of cell septation and division and maintains the inner membrane integrity (Pal et al. 2019). The

data of total lipid analysis reveals inhibited Lyso-PIM3, Ac1PIM1, PIM2, PIM1 in Δ disA (Fig. 4f). Similarly, the cell wall lipidome also showed inhibited PG and PI in Δ disA (Fig. 5f). Altogether, these observations reinforce that deletion of RecA gene changes the composition of *Mycobacterium* membrane by affecting PG, PE, Lyso-PIM1, and PIMs leading to disrupted membrane integrity. Further, we performed a comparative analysis of lipid categories identified from the MycoMass database with MS-LAMP's output. The search in MycoMass database was also done in the same manner as the search was done with Mtb LipidDB using MS-LAMP, whereby the window range was set to 1.0. Therefore, when the outputs corresponding to every lipid category resulting from each of these two databases were compared, we observed similar trends (Fig. S6).

A clinically relevant virulence trait of MTB is its ability to form biofilms, which are resistant to anti-TB drugs. Biofilm development is an essential factor for MDR and pathogenicity of MTB (Brennan et. al. 2017). We explored that biofilm formation was considerably reduced in Δ disA, as revealed by CV staining and quantification of metabolic activity and biofilm biomass (Fig. 6). We further analyzed the transcripts of genes associated with DNA damage, efflux activity, membrane integrity, ROS and biofilm formation to support our biochemical assays (Fig. 7; S7). *C. elegans* has become an extensively used model organism for studying immunology and host-pathogen interactions. Hence, the in-vivo studies with nematode model *C. elegans* confirmed that infected worms survival rate increased in presence Δ disA as compared to WT (Fig. 8) which further verified Δ disA as a potential drug target.

Conclusion

Taken together, the data presented in the study depicted indispensability of RecA for functional cell membrane integrity, drug efflux and biofilm formation in *M. smegmatis*, but the mechanisms that underlie this crosstalk remain to be defined. Dissecting these mechanisms may hold promise for revealing new therapeutic targets for life-threatening TB infection.

Acknowledgement

We are grateful to Krishna Murari Sinha, for providing *M. smegmatis* RecA (Δ disA) deletion mutant as generous gift. We thank Anindya Ghosh, for providing wild-type *C. elegans* (N2) and *Escherichia coli* OP50 strains. We thank Amity Lipidomics Research Facility (ALRF), Amity University Haryana for assisting us in mass spectrometry experiments.

Conflict of Interests

None to declare

Supporting Information

Table S1. List of primers used for RT-PCR.

Figure S1. Growth curve of WT and Δ disA.

Figure S2-S5. Representative chromatogram and mass spectra obtained from LC-MS of WT and Δ disA total and cell wall lipids.

Fig S6: Comparison of the output obtained from MycoMass with Mtb LipidDB (MS-LAMP) for the WT and Δ disA.

Fig S7: Raw data of RT-PCR of WT and Δ disA .

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Figure Legends

Fig. 1 Effect of RecA deletion on RIF susceptibility. Broth micro dilution assay to determine the MIC₉₀ of WT and Δ disA in presence of RIFMIC is represented as optical density on y axis and drug concentration on x- axis in logarithmic scale. Data represents that the mean O.D₆₀₀ of three independently completed experiments.

Fig. 2 Effect of RecA deletion on oxidative stress. (a) ROS generation with DCFDA as observed by fluorescence microscope in WT and Δ disA. Scale bar depicts 10 μ m. (b) ROS quantification in the presence of INT dye in WT and Δ disA. Mean of O.D₄₉₀ $\pm$ SD of three independent sets of experiments are depicted on Y-axis and * depicts *P* value 0.042. (c) The malondialdehyde (MDA) content was determined using a reaction with thiobarbituric acid (TBA) to estimate lipid peroxidation. Mean of O.D₅₃₅ nm $\pm$ SD of three independent sets of experiments are depicted on Y-axis and * depicts *P* value 0.0026.

Fig.3 Effect of RecA deletion on membrane integrity, cell surface properties and efflux pump. (a) Spot assay of Δ disA and WT with membrane perturbing agent SDS. (b) Fluorescence microscopy showing of PI staining (red signal) of Δ disA and WT cells at 40x magnification. Scale bar depicts 10 μ m (c) Nitrocefin membrane permeability assay of Δ disA and WT. Mean of O.D₄₈₅ $\pm$ SD of three independent sets are depicted on y axis, *P* value 0.0049. (d) Colony morphology of Δ disA and WT were observed on 7H10-based medium at 10x magnification. Scale bar depicts 20 μ m (e) Sliding motility of Δ disA and WT. (f) Cell sedimentation of Δ disA and WT. Left panel shows O.D₆₀₀ of WT and Δ disA cells depicted on y axis with respect to time (hours) on x-axis, *P* value 0.048. Right panel shows sedimentation rates per hour on y-axis of WT and Δ disA cells, calculated by estimating the difference in O.D₆₀₀ from 0 till 22 h per unit

time interval. Mean of three independent sets of experiments SD (shown by error bars) are depicted, P value 0.043. (g) Efflux pump activity is estimated by measuring extracellular concentrations of EtBr in WT and Δ disA. Negative control represents de-energized cells without glucose. Mean of O.D₅₂₇ SD of three independent sets of experiments are depicted on y-axis with respect to time (minutes) on x-axis, P value 0.032.

Fig 4. Effect of RecA deletion on total lipidome. (a) The Bar graph depicts number of observed m/z values at 1 WR in “Total lipids” of WT and Δ disA identified from Mtb LipidDB (MS-LAMP). (b) Venn diagram depicts number of common lipid moieties in WT and Δ disA identified from m/z values at 1 WR through Mtb LipidDB (MS-LAMP). (c) Distribution of number of lipids in each class at 1WR in WT and Δ disA. (d). Bar graph depicts the number of FA subclasses identified from m/z values at WR 1 from total lipid in WT and Δ disA. (e) Bar graph depicts the number of GL subclasses identified from m/z values at WR 1 from total lipid in WT and Δ disA. (f) Bar graph depicts the number of GPL subclasses identified from m/z values at WR 1 from total lipid in WT and Δ disA.

Fig 5. Effect of RecA deletion on cell wall lipidome. (a) The Bar graph depicts number of observed m/z values at 1 WR in “cell wall lipids” of WT and Δ disA identified from Mtb LipidDB (MS-LAMP). (b) Venn diagram depicts number of common lipid moieties in WT and Δ disA identified from m/z values at 1 WR through Mtb LipidDB (MS-LAMP). (c) Distribution of number of lipids in each class at 1WR in WT and Δ disA. (d). Bar graph depicts the number of FA subclasses identified from m/z values at WR 1 from cell wall lipid in WT and Δ disA. (e) Bar graph depicts the number of GL subclasses identified from m/z values at WR 1 from cell wall

lipid in WT and Δ disA. **(f)** Bar graph depicts the number of GPL subclasses identified from m/z values at WR 1 from cell wall lipid in WT and Δ disA.

Fig.6 Effect of RecA deletion on biofilm formation. (a). The left panel show CV staining depicting biofilm formation in WT and Δ disA. Scale bar depicts 20 μ m. **(b)** Metabolic activity (expressed as O.D₄₅₀) depicted as a bar graph. Data are expressed as mean $\pm$ SD of three independent sets of experiments. * Depicts *P* value 0.0026. **(c)** Biofilm biomass (dry weight) in WT and Δ disA. Mean of dry weight $\pm$ SD of three independent experiments are depicted on the Y-axis and * depicts *P* value 0.015.

Fig.7 RT-PCR of differentially regulated genes in Δ disA. The left panel show transcript levels of *kat g*, *aftd*, *pe*, *accd4* and *lig a* in lanes WT and Δ disA strains. The right side panel shows the quantitation (density expressed as Intensity/mm²) of the respective transcripts normalized with constitutively expressed 16s transcript (Fig. S7).

Fig.8 Effect of RecA deletion on the survival of *C. elegans*. Worms were scored every day for survival and worms that ruptured were removed from the study. Kaplan–Meier graph depicting survival of *C. elegans* on infection with WT and Δ disA strains. Data are expressed as mean of three independent sets of experiments (*P* value 0.006).

Fig.1

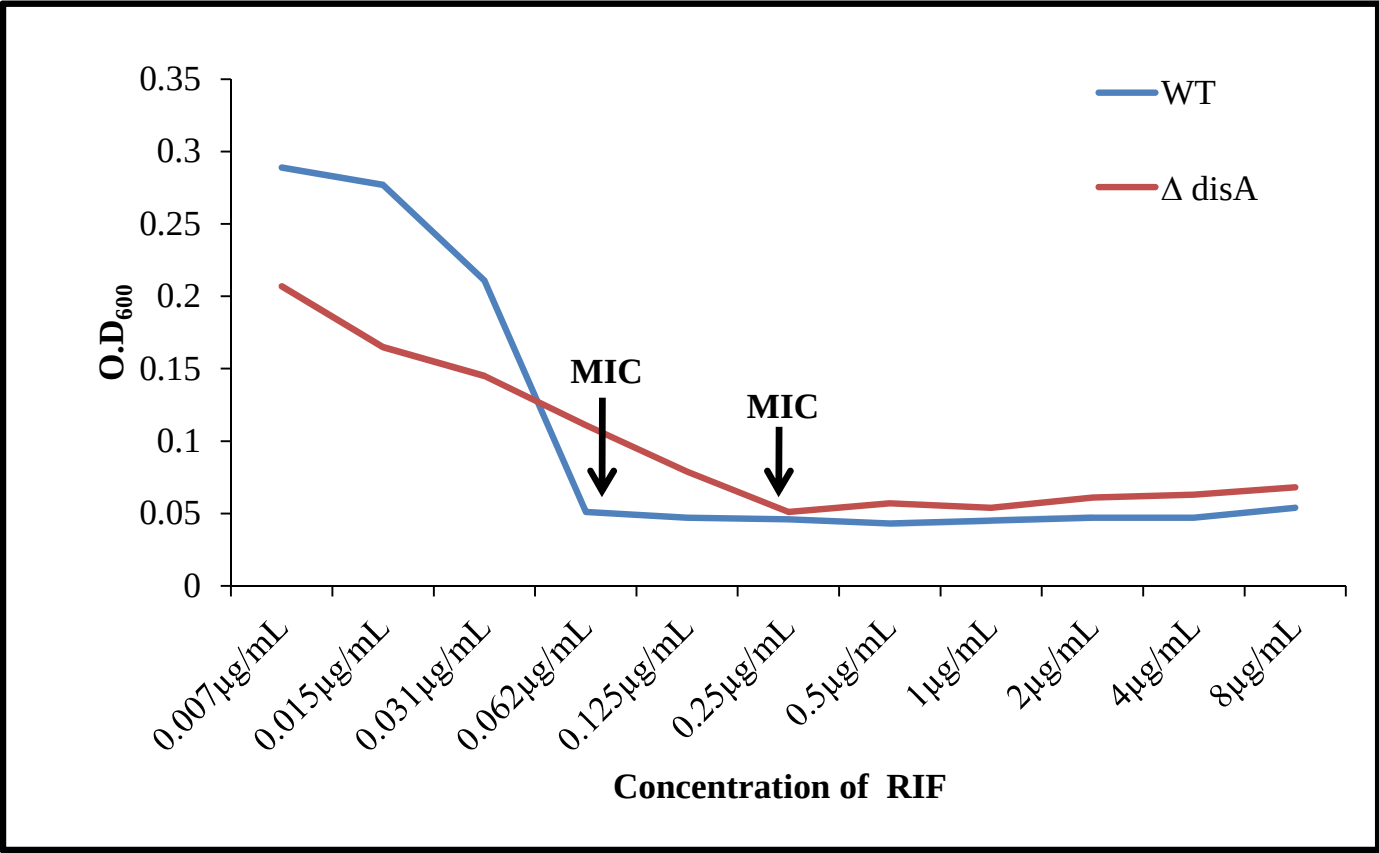


Fig.2

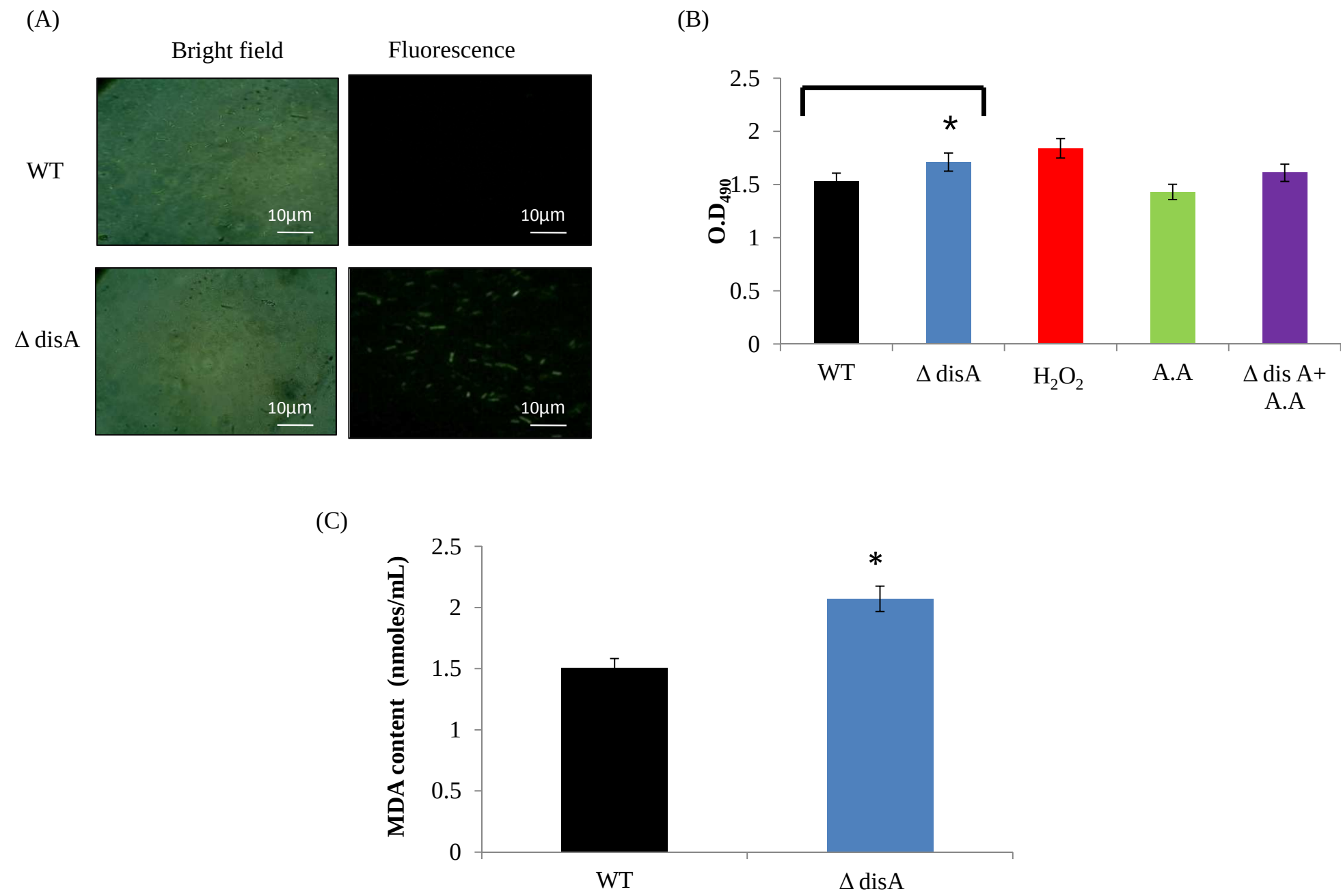


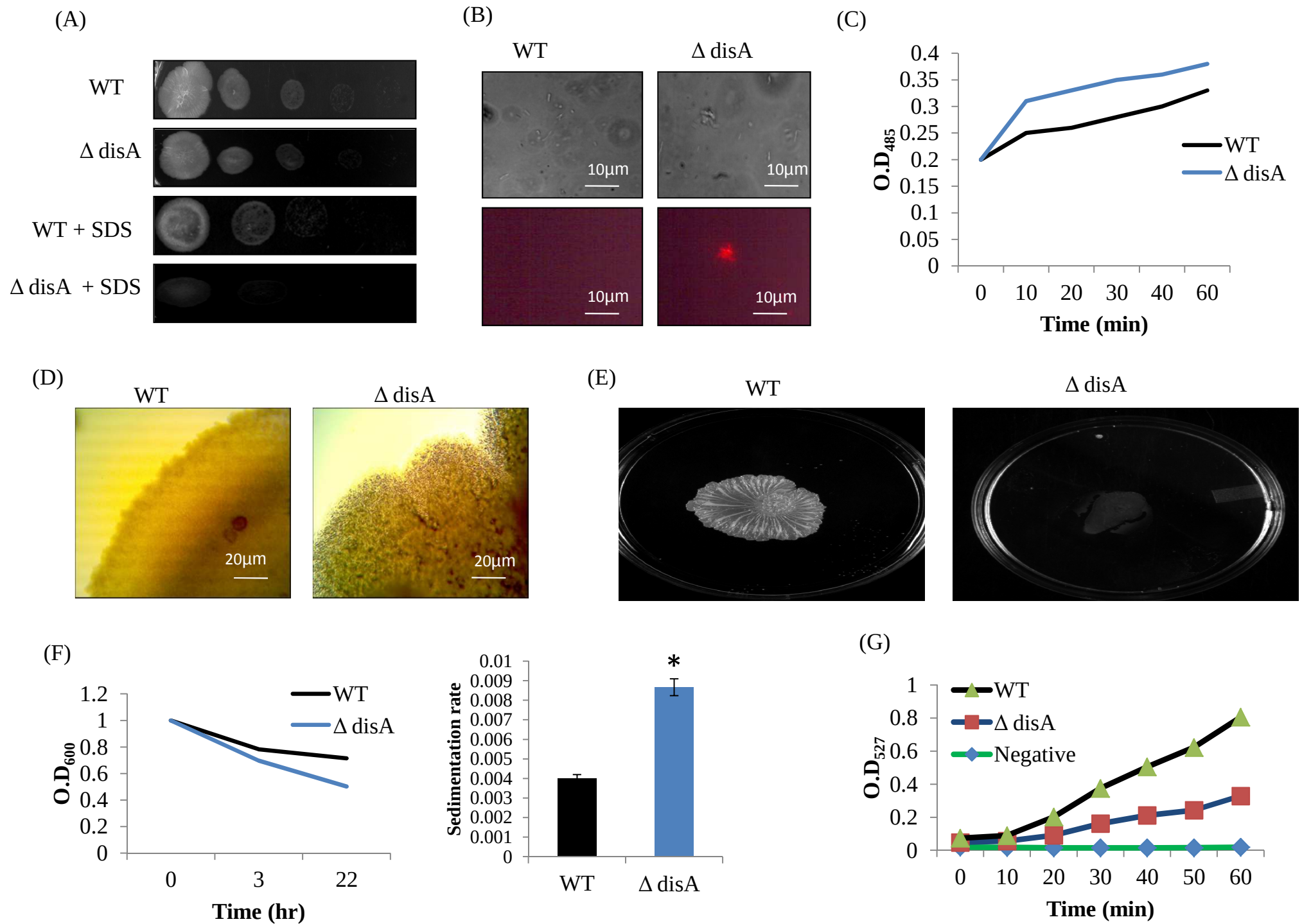
Fig.3

Fig. 4

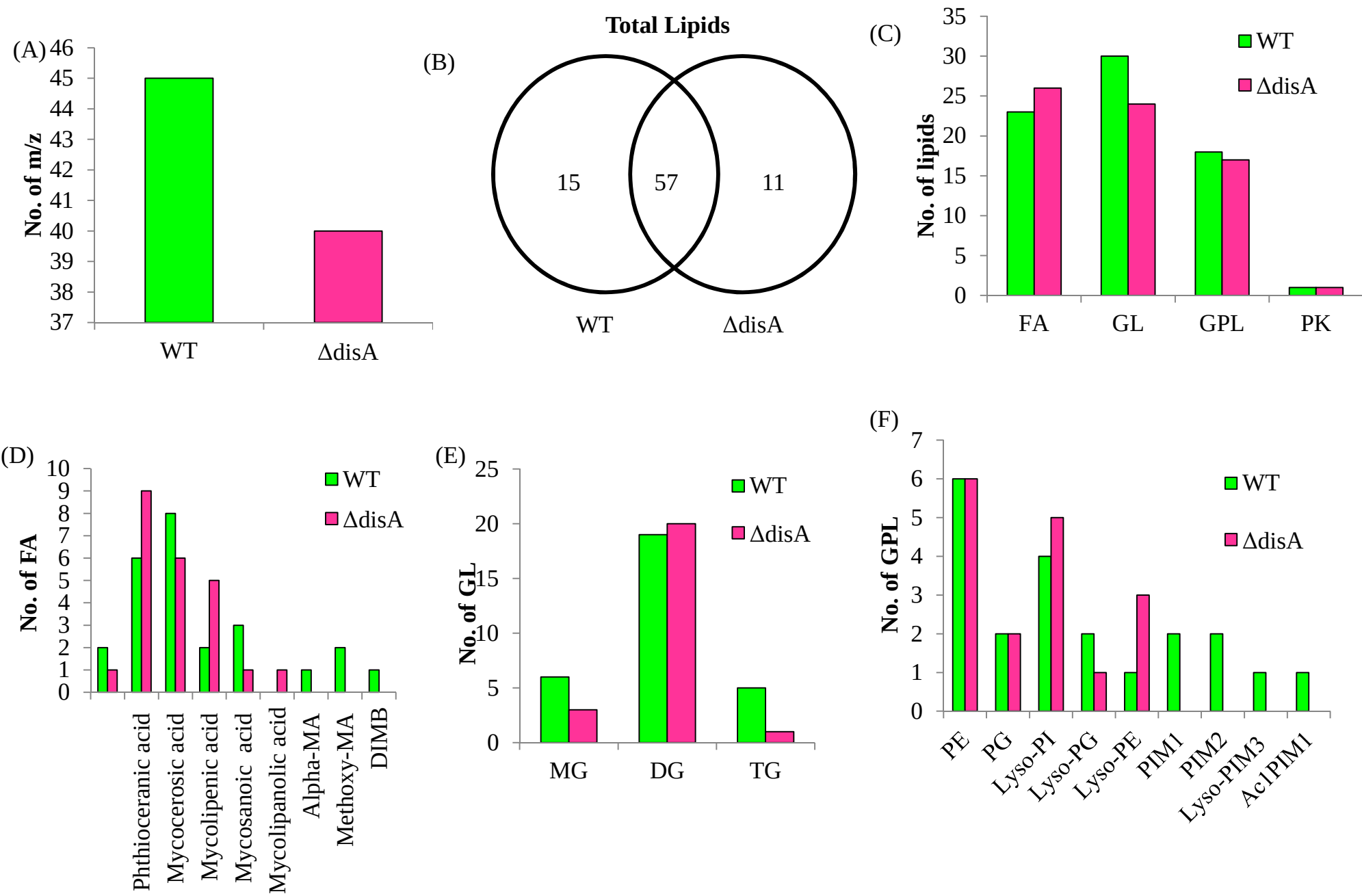


Fig. 5

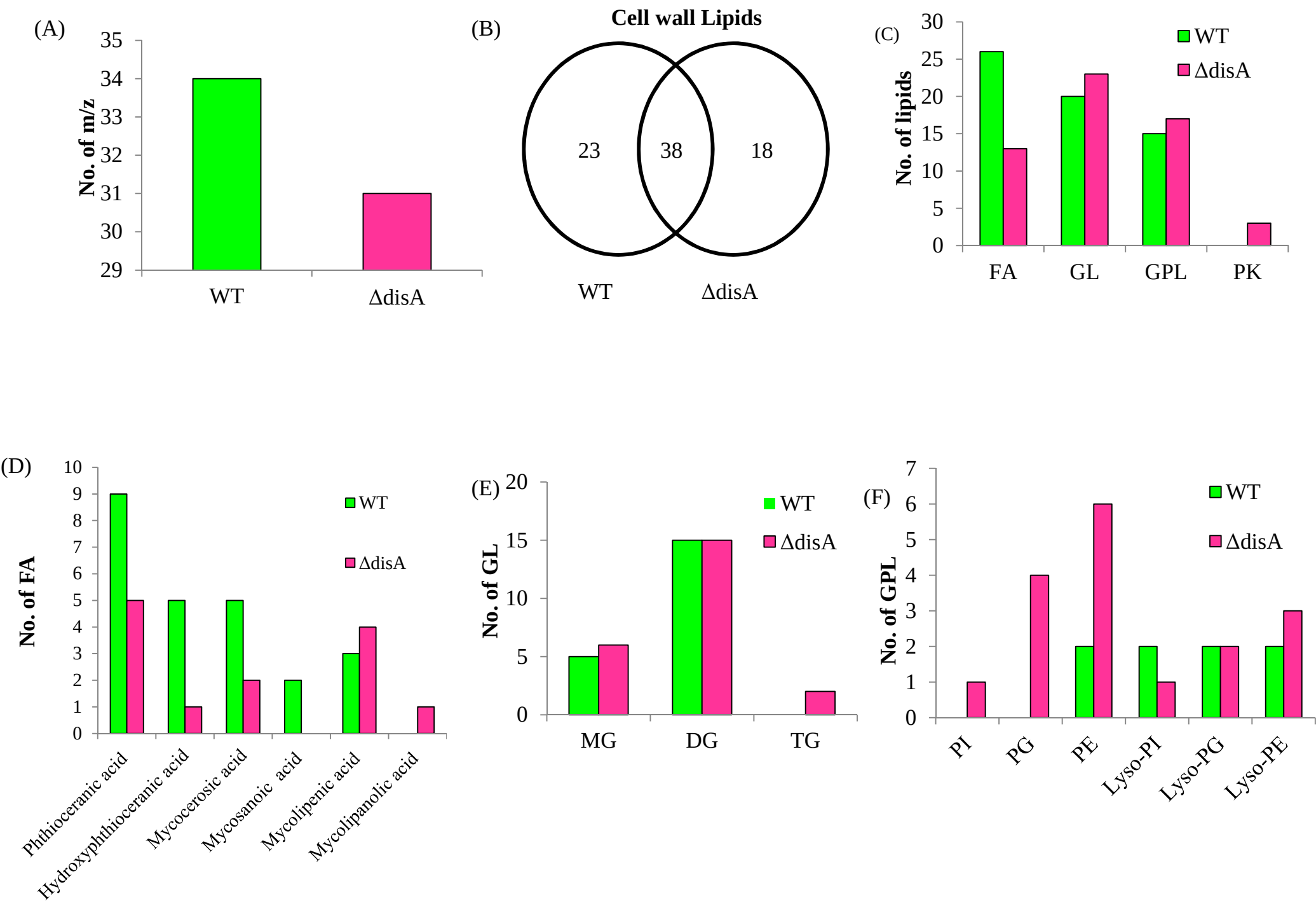


Fig. 6

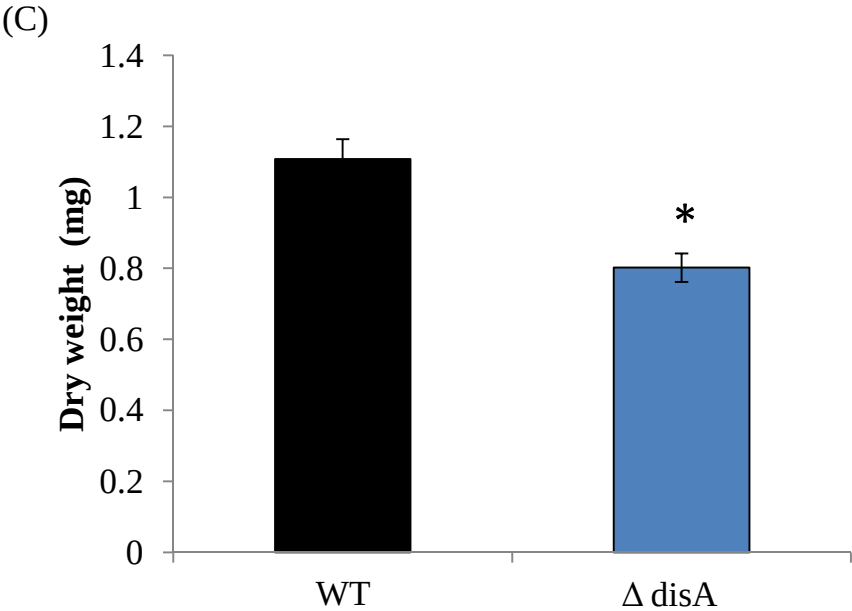
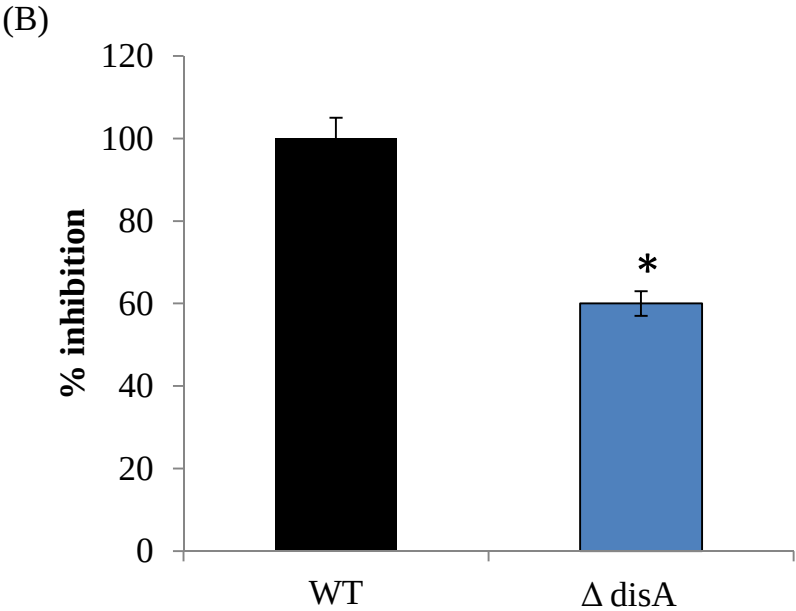
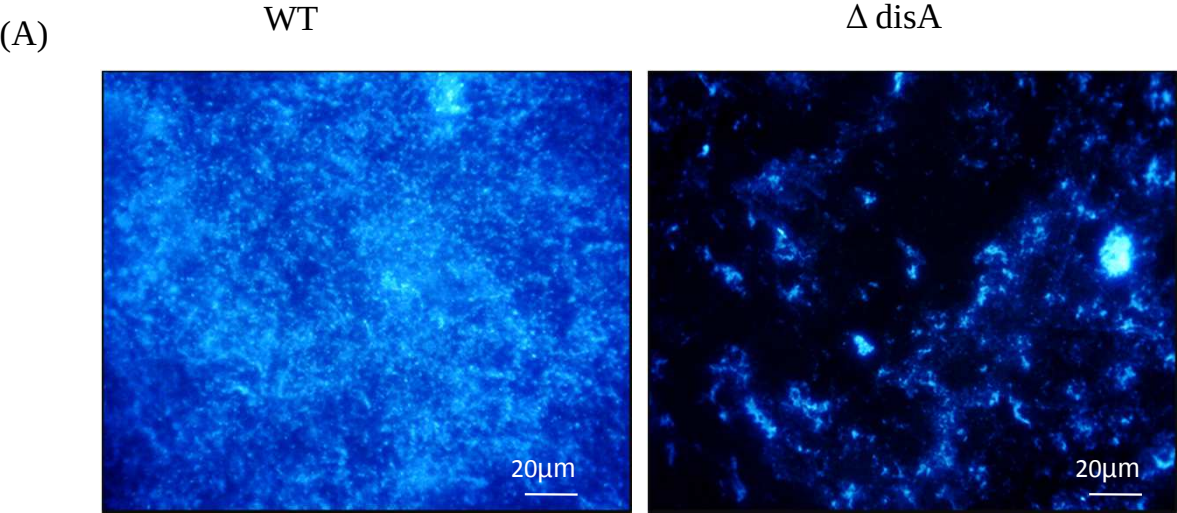


Fig. 7

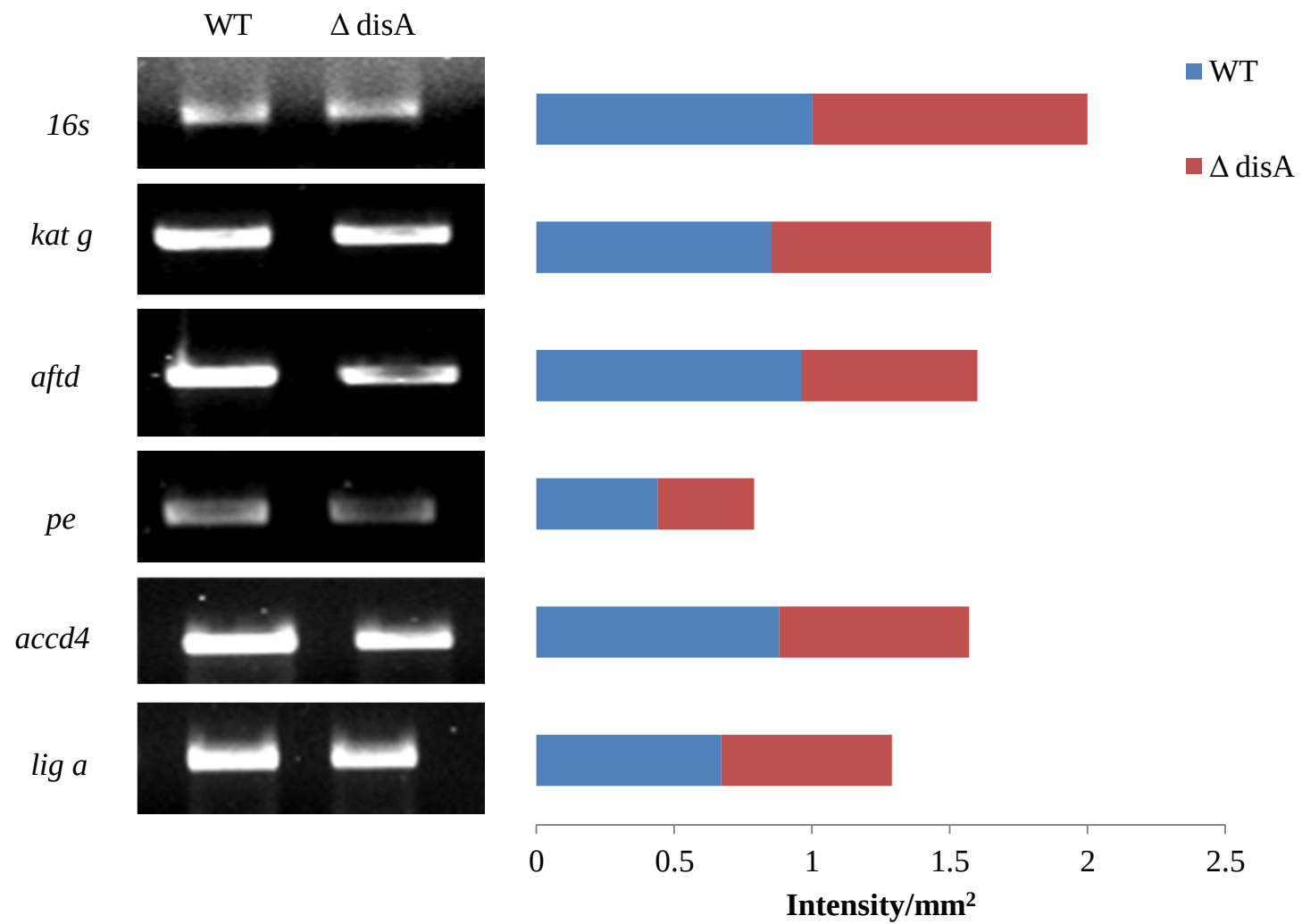
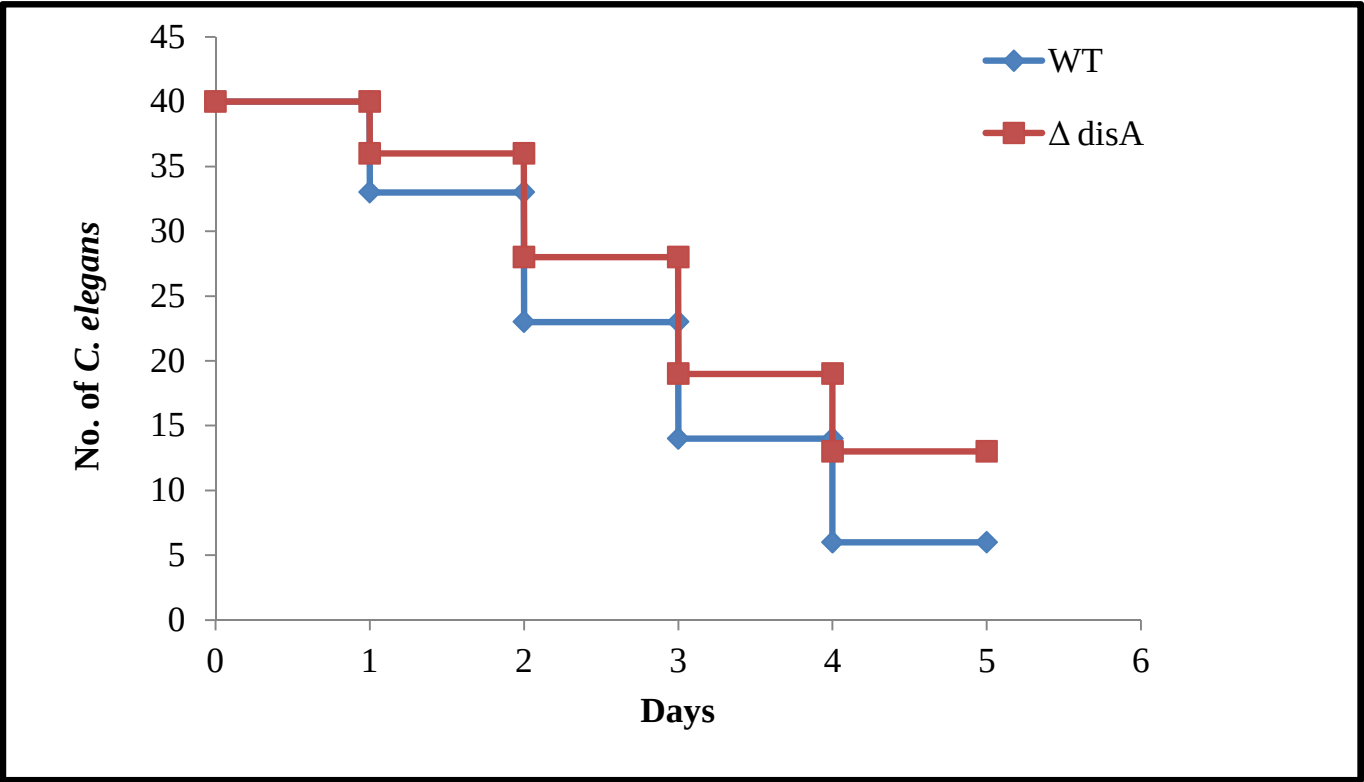


Fig. 8



Highlights

- Rifampicin susceptibility is enhanced in RecA deletion mutant (Δ disA).
- Endogenous ROS levels are enhanced in Δ disA.
- Lipid peroxidation is induced in Δ disA.
- Membrane integrity is disrupted in Δ disA.
- Cell surface phenotypes are altered in Δ disA.
- Efflux pump activity is impeded in Δ disA.
- Lipid profile is altered in Δ disA.
- Biofilm formation is inhibited in Δ disA.
- *C. elegans* infectivity is reduced with Δ disA.

Conflict of Interest

Declarations of interest: None